# Online Appendix

Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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## OA.1 Algorithms

**Algorithm OA.1 (amortized shape model).** (i) Assemble the pooled panel $\{(s_{j,t}, \widehat{z}_{j,t})\}$ over names $j$ and days $t$. (ii) Train $H_\phi$ by SGD on $\mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z} - H_\phi(s,\tau))$ [network], or boost per-$\tau$ regressions on a grid [trees]; enforce monotonicity by rearrangement. (iii) For any asset — including one with no return history, using characteristics-only $s$ — evaluate $H_\phi(s_t, \cdot)$; no per-asset estimation step exists. (iv) To *sample*, draw $\tau^{(m)} \sim U(0,1)$ and return $z^{(m)} = H_\phi(s_t, \tau^{(m)})$.

**Algorithm OA.2 (the misspecification meter, as deployed).** (i) Fit equation (1) of the paper on a trailing window; store $\widehat{z}_{t-62}, \ldots, \widehat{z}_t$. (ii) Compute $\mathrm{mk}_{63}$, $|\mathrm{sk}_{63}|$, $J_5$ by equation (12) of the paper. (iii) Convert each to a decile rank against the trailing pooled panel (same asset class); the score is the rank of the maximum. (iv) Act by the paper: top decile $\Rightarrow$ the parametric model's average pinball loss runs ∼2–3% higher within this score region (a conditional average, not a per-name point estimate) — flag the asset for the nonparametric / residual-hybrid engine and for model-risk review; elsewhere $\Rightarrow$ no action (in the amortized setting the engine is run regardless; the meter serves surveillance, and day-level switching is unsupported, Section 5.4 of the paper). (v) Latency: detection lags episode onset by median 0 / mean 2–4 days; the edge is back-loaded within episodes, so the lag forfeits little.

**The production loop.** *Nightly:* (i) update each name's GARCH filter state $\widehat{\sigma}_{t+1}$ (recursion only; parameters refit annually); (ii) assemble the state vector $s_t$ and evaluate the shared shape model $q_\phi(\tau \mid s_t)$ — one model for the whole book, no per-name estimation; (iii) apply the frozen GPD tail below $p_0$ and, where the desk elects the coverage overlay, the conformal shift $c_\tau$ (static or adaptive); (iv) report $\widehat{Q}_{t+1}(\tau) = \widehat{\sigma}_{t+1}\,\widetilde{q}(\tau \mid s_t)$ at the desk's levels, with $\mathrm{ES}_{97.5}$ from the GPD closed form below the splice; (v) read the misspecification score equation (12) of the paper as the model-risk monitor — top decile means the parametric fallback's conditional average loss in that score region runs ~2–3% higher. *Annually:* refit filters, retrain the shape model, re-estimate tail and shifts, all on trailing data. *New listing:* run the separate characteristics-only amortized forecaster until an own-history sample sufficient to initialize the per-name scale filter accrues; then attach this residual-hybrid pipeline. Total marginal cost per name per day: one filter update and one model evaluation.

## OA.2 Supplementary figures

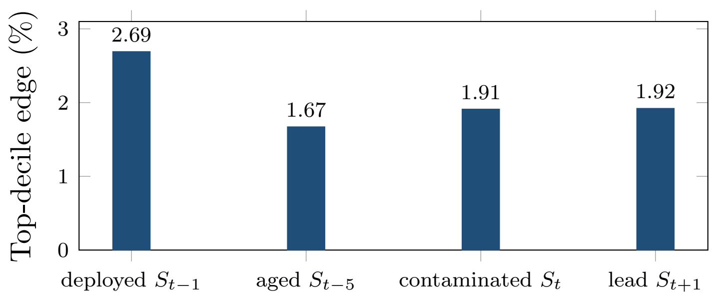



**Figure OA.1:** Timing diagnostics for the frontier. Top-decile edge under four alignments of the $\mathrm{mk}_{63}$ score with the daily loss (200 CRSP names; per-date DM 5.5–6.5 in all four). The deployed lag dominates, and the contaminated same-day alignment yields less — the opposite of what a look-ahead artifact would produce.

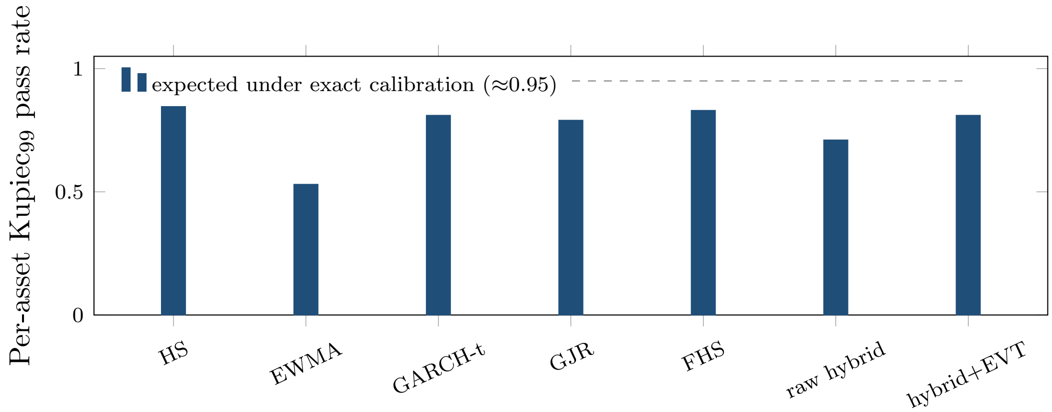



**Figure OA.2:** Exception tests restated per asset (140 CRSP names, 99% VaR): share of names whose individual Kupiec test is consistent with nominal. HS over-covers (conservative); EWMA fails; the EVT-tailed engine matches the parametric benchmarks while beating them on loss and the joint FZ0 score (Table 6 of the paper). Date-clustered tests in the table notes. The GJR bar predates the skew-$t$ correction of Table 6's note; the corrected pooled battery row passes both 99% tests.

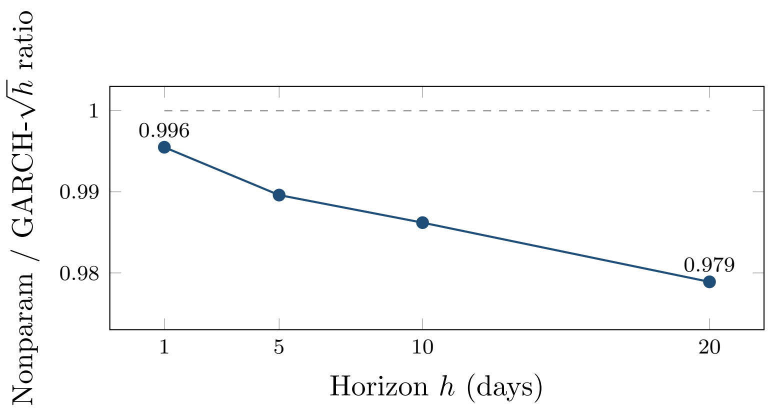
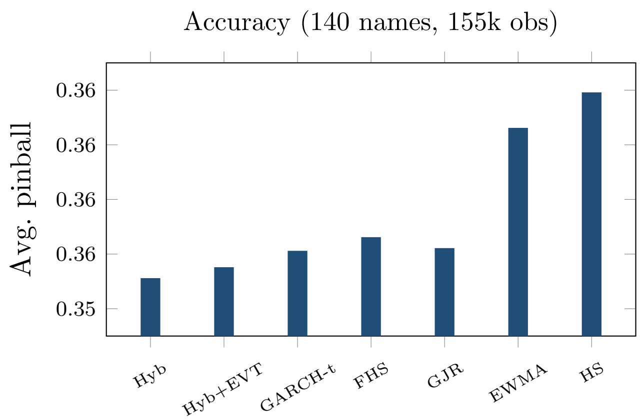



**Figure OA.3:** The edge grows with horizon. Direct-model/GARCH pinball ratio at $h = 1, 5, 10, 20$ days (150-name design panel; training labels whose $h$-day window crosses the split are purged): fat tails compound while $\sqrt{h}$ scaling compounds the shape error — 0.45% at $h$=1 to 2.1% at $h$=20. The forecaster is the direct multi-day model of the main text's ten-day subsection, *not* the residual-hybrid engine, whose architecture is one-day. An earlier version of this figure showed larger ratios from an unpurged run; the purged rerun supersedes it.



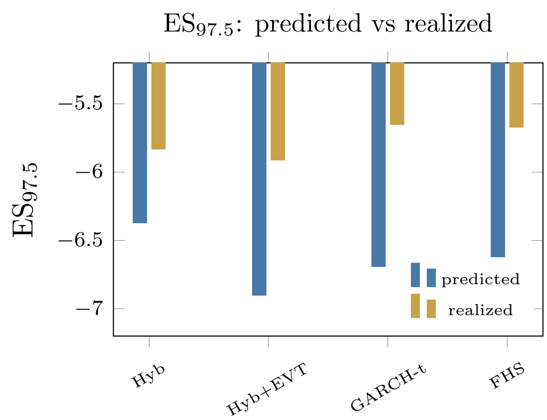




**Figure OA.4:** The FRTB battery. Left: average pinball loss by model (axis truncated to resolve the ordering; differences carry DM 5.2–8.8 vs the hybrid, all $p \approx 0$); CAViaR, not shown, statistically ties the hybrid. Right: $\mathrm{ES}_{97.5}$ predicted (exact tail integral) vs realized below each model's own VaR — a model-dependent diagnostic, not a ranking (every fat-tailed entrant's integral exceeds its own-tail mean); the joint FZ0 score of the paper is the ranking criterion.

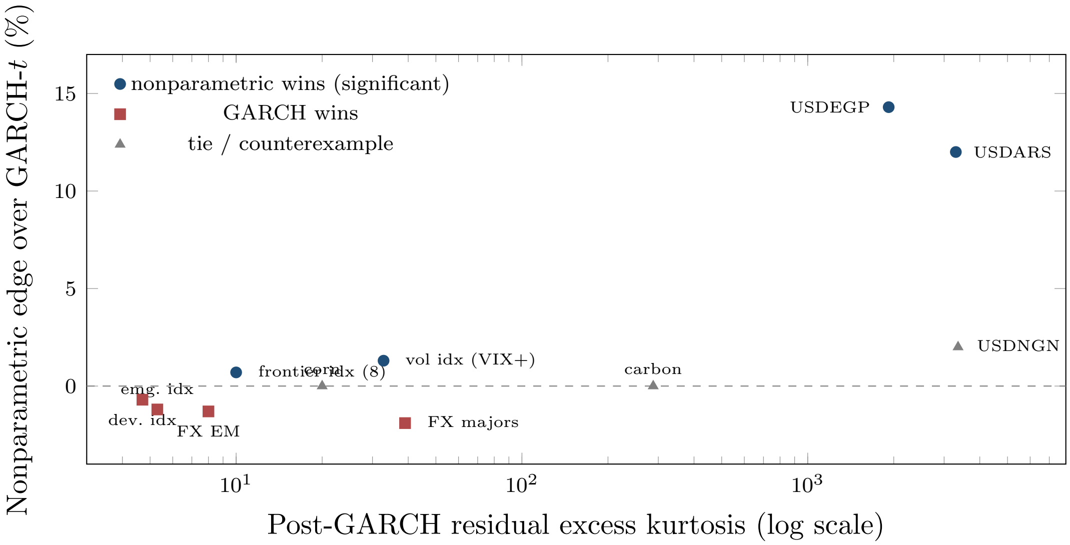



**Figure OA.5:** The frontier across universes: nonparametric edge vs post-GARCH residual kurtosis, tier means and single assets with study-reported kurtosis only (country universe corr 0.53; cross-asset corr 0.43). The hyperinflation-FX tier (kurtosis ∼2000–3300) carries the only decisive standalone wins; high kurtosis is necessary but not sufficient — carbon (288) and corn (20) tie, and NGN's win is not significant. The Korea-2026 control (kurtosis 4–13, GARCH wins all 23 assets) sits in the lower-left regime.

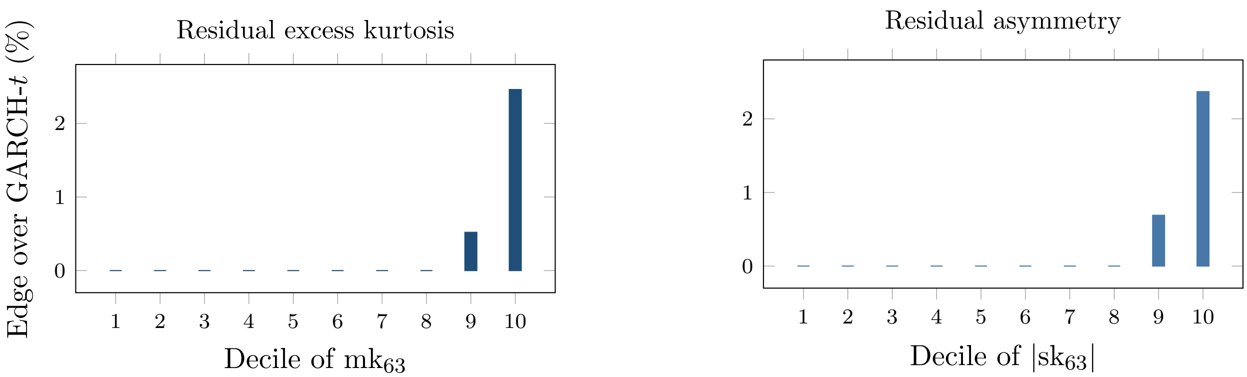



**Figure OA.6:** The misspecification frontier. Nonparametric pinball edge over GARCH-$t$ by decile of the trailing 63-day residual excess kurtosis (left) and residual asymmetry (right), 200 CRSP names, 221.6k test asset-days. Deciles 1–8 are $\approx 0$ (drawn at zero; exact bulk values in the result file); the edge jumps in the top decile (+2.46% kurtosis, +2.37% asymmetry; per-date DM 6.5 in the top kurtosis decile).

## OA.3 Supplementary tables

**Table OA.1:** Double sort: the edge needs both a shape break and an active regime. Cells are the nonparametric-over-GARCH-$t$ pinball edge (%) in independent quintiles of the misspecification score (rows) and trailing volatility (columns), design-era test panel.

| | $\text{rv}_{63}$ Q1 (calm) | Q2 | Q3 | Q4 | Q5 (active) |
|---|---|---|---|---|---|
| $\text{mk}_{63}$ Q1 (low) | +0.24 | +0.34 | +0.10 | +0.01 | −0.14 |
| Q2 | +0.16 | +0.20 | +0.22 | +0.09 | −0.57 |
| Q3 | −0.02 | +0.19 | +0.25 | +0.44 | +0.04 |
| Q4 | +0.08 | +0.15 | +0.35 | +0.23 | +0.10 |
| Q5 (high) | −0.10 | +0.44 | +0.68 | +0.61 | +**2.33** |

Top-row date-clustered DM by volatility quintile: 0.7, 2.3, 1.2, 2.8, 6.4. The bottom-left and top-right regions are the point: neither ingredient alone produces the concentration.

**Table OA.2:** Model selection by task and regime. Each row is established in the section cited.

| Task / regime | Winner | Where |
|---|---|---|
| Point quantile accuracy, tabular state | GBM (trees) | Section 3.2 of the paper |
| Single asset, calm (∼90% of asset-days) | GARCH-$t$ | Section 5 of the paper |
| Single asset, top misspecification decile | Nonparametric (+2.7%, DM 6.5) | Section 5 of the paper |
| Amortized cross-name panel, any regime | Nonparametric always; no gate | Section 5.4 of the paper |
| Regulatory ES with a ≥GARCH floor | Residual-hybrid + EVT + conformal | Section 6 of the paper |

**Table OA.3:** Cross-universe multiplicity: Holm step-down over all 93 instrument-level comparisons in the three exploratory universes of the paper's cross-universe table (24 FX pairs, 26 country indices, 43 cross-asset instruments). One-sided $p$ from each instrument's date-clustered DM statistic; familywise $\alpha = 5\%$. Ranks below twelve all have $p > 0.02$ and are omitted.

| Rank | Instrument | Universe | DM | $p$ (one-sided) | Holm cutoff | Verdict |
|---|---|---|---|---|---|---|
| 1 | PJM power[a] | cross-asset | 283.1 | $< 10^{-16}$ | $5.38 \times 10^{-4}$ | artifact[a] |
| 2 | Baltic Dry | cross-asset | 9.36 | $< 10^{-16}$ | $5.43 \times 10^{-4}$ | survives |
| 3 | VIX | cross-asset | 5.71 | $5.7 \times 10^{-9}$ | $5.49 \times 10^{-4}$ | survives |
| 4 | V2X | cross-asset | 5.03 | $2.5 \times 10^{-7}$ | $5.56 \times 10^{-4}$ | survives |
| 5 | VVIX | cross-asset | 4.66 | $1.6 \times 10^{-6}$ | $5.62 \times 10^{-4}$ | survives |
| 6 | USDEGP | FX | 3.80 | $7.2 \times 10^{-5}$ | $5.68 \times 10^{-4}$ | survives |
| 7 | Kenya NSE20 | indices | 3.52 | $2.2 \times 10^{-4}$ | $5.75 \times 10^{-4}$ | survives |
| 8 | Wheat | cross-asset | 3.51 | $2.2 \times 10^{-4}$ | $5.81 \times 10^{-4}$ | survives |
| 9 | MOVE | cross-asset | 3.22 | $6.4 \times 10^{-4}$ | $5.88 \times 10^{-4}$ | fails |
| 10 | Sri Lanka CSEALL | indices | 3.20 | $6.9 \times 10^{-4}$ | $5.95 \times 10^{-4}$ | fails |
| 11 | Gasoline | cross-asset | 3.20 | $6.9 \times 10^{-4}$ | $6.02 \times 10^{-4}$ | fails |
| 12 | USDARS | FX | 2.86 | $2.1 \times 10^{-3}$ | $6.10 \times 10^{-4}$ | fails |

The family is every instrument-level look in the three exploratory universes; the Korea-2026 negative control (23 assets, GARCH-$t$ wins throughout) is not part of the win family. Bonferroni at 0.05/93 gives the identical survivor set. The pooled crisis-FX comparison (DM 4.12, $p \approx 1.9 \times 10^{-5}$) and the kurtosis–edge correlations (0.53 across indices, 0.43 cross-asset) are pooled and gradient statistics outside this instrument-level family; they are the replication evidence the paper's cross-universe claims rest on.

[a] Retained in the family because the raw sweep contains it, but not a win: the paper's cross-asset audit reclassifies electricity as a log-return artifact on near-zero and negative prices — refit on price differences, GARCH wins PJM (DM −2.73) — so the survivor count the text quotes is seven.

## OA.4 The two-regime example

### A synthetic boundary check: when the frontier does *not* appear

Before the real data, a simulation with known truth (run once, seed fixed) checks the formal results and, more usefully, delineates when the frontier edge should *fail* to appear. The data-generating process is deliberately favorable to the thermometer: unit-variance Student-$t$ residuals whose degrees of freedom switch between a calm regime ($\nu = 12$) and a fat regime ($\nu = 3$) under a persistent Markov chain; forecasters see the trailing 63-day kurtosis score and compete at $\tau \in \{0.01, 0.025\}$ (80,000 days; scale known, isolating shape). Three results. First, the Lemma 1 of the paper identity verifies numerically: the regret integral and the Monte-Carlo regret agree to five decimals ($3.5 \times 10^{-5}$ at $\tau = 0.01$). Second, the conformal shift moves realized coverage toward nominal on the held-out split, as Proposition 1 of the paper promises. Third, and most instructive, the score-binned nonparametric forecaster does *not* beat the fixed shape here: a single $t_{\widehat{\nu}}$ fitted on the mixture ($\widehat{\nu} = 6.8$) is nearly unbeatable at these levels, with total regret under 0.5% and no decile gradient. Two mechanisms explain it, and both teach. (i) By Lemma 1 of the paper regret is *quadratic* in the quantile wedge, and at $\tau \geq 0.01$ the wedge between the fitted compromise and either regime is small; static kurtosis *blending* is a second-order sin. (ii) The sample kurtosis of a $t_\nu$ law has infinite estimator variance for $\nu \leq 8$, so a moment-based thermometer is at its noisiest just when tails are fat, and sixty-three observations cannot reliably rank regimes that differ only in a static $\nu$. In the empirical sorts this noise works *against* the frontier, not for it: misclassification across bucket boundaries attenuates measured top-bucket edges, so the reported concentration is if anything understated; the jump and asymmetry signals, whose top deciles independently survive familywise control, corroborate the ordering, and a tail-index (Hill-type) variant of the score is an obvious robustness extension. For the empirical sections this means the real-data top-decile edge of Section 5 of the paper cannot be an artifact of stationary fat tails being averaged over (that configuration is the one the simulation shows a fixed shape handles) but must reflect *local* shape breaks a single fitted $\nu$ cannot straddle: asymmetry, jump clustering, and residual misfit that co-move with the score. This is the same conclusion the Korea 2026 negative control reaches from the opposite direction (price crashes are not residual misspecification), and it is why the score combines kurtosis with asymmetry and a jump indicator rather than kurtosis alone. (Simulation script `toy_example.py` ships with the code release.)

## OA.5 Trading implications and scenario generation

**Trading implications.** The universes where the nonparametric tail wins — crisis FX, volatility indices, credit spreads, frontier equities — are the desks where tail-shape-aware sizing and hedging matter most. The *direction* of the trade matters as much as its location: a correct physicalmeasure tail forecast does not by itself make bought crash insurance profitable, because the crash risk premium can survive conditioning on a correct forecast. The deployable side is the other one — *warehousing* (selling) the premium — with the model supplying tail control rather than alpha on the mean: model-timed overlays improve tail outcomes, not average returns, consistent with the tail being the only place the model knows something a fixed-shape filter does not. The contribution to a premium-harvesting book is risk management (when to size down or step aside ahead of a genuine crash), not signal generation.

**Scenario generation.** Because the shape stage carries an exact inverse-CDF sampler, calibrating intractable scenario generators (rough volatility, Hawkes, jump-diffusions) for CCAR/ICAAP-style stress design is a natural extension; it is not developed here.

## OA.6 Proofs

## OA.7 Proofs

The four formal results of Section 3.3 of the paper are stated with proof sketches in the body; the complete arguments, with all regularity hypotheses made explicit, are collected here. Throughout, $\rho_\tau(u) = u(\tau - \mathbb{I}\{u < 0\})$ and $q_F = F^{-1}(\tau) = \inf\{x : F(x) \geq \tau\}$ is the lower generalized inverse.

### Proof of Lemma 1 of the paper (pinball regret identity)

Assume $\mathbb{E}|Z| < \infty$, so

$$S(q) = (1-\tau)\int_{(-\infty,q]} (q-z)\, dF(z) \;+\; \tau \int_{(q,\infty)} (z-q)\, dF(z)$$

is finite for every $q$, and $S$ is convex (an expectation of the convex maps $q \mapsto \rho_\tau(z-q)$). For fixed $z$, $q \mapsto \rho_\tau(z-q)$ is Lipschitz with constant $\max(\tau, 1-\tau)$, so the difference quotients $[\rho_\tau(Z-q') - \rho_\tau(Z-q)]/(q'-q)$ are bounded by the integrable constant $\max(\tau, 1-\tau)$; dominated convergence justifies differentiating under the expectation wherever the one-sided limits exist. Off the atoms of $F$,

$$\frac{d}{dq}\,\rho_\tau(z-q) = \mathbb{I}\{z < q\} - \tau, \qquad \Longrightarrow \qquad S'(q) = \mathbb{E}\,\mathbb{I}\{Z < q\} - \tau = F(q^-) - \tau,$$

which equals $F(q) - \tau$ for all but countably many $q$. Since $S$ is convex, hence absolutely continuous, the fundamental theorem of calculus gives

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\, du,$$

the countable atom set being Lebesgue-null; *no continuity hypothesis on $F$ is needed for this identity.* For nonnegativity, note $q_F = \inf\{x : F(x) \geq \tau\}$ implies $F(u) \geq \tau$ for $u \geq q_F$ and $F(u) < \tau$ for $u < q_F$. If $q > q_F$ the integrand is $\geq 0$ on $[q_F, q]$, so the integral is $\geq 0$; if $q < q_F$,

$$S(q) - S(q_F) = \int_{q_F}^{q} \big(F(u) - \tau\big)\, du = \int_{q}^{q_F} \big(\tau - F(u)\big)\, du \;\geq\; 0,$$

the integrand again $\geq 0$ on $[q, q_F]$. Equality holds iff $F(u) = \tau$ for Lebesgue-a.e. $u$ between $q_F$ and $q$. Finally, if $F$ is absolutely continuous with density $0 < m \leq f \leq M$ on the interval joining $q_F$ and $q$, then $F$ is continuous there, so $F(q_F) = \tau$ and $F(u) - \tau = \int_{q_F}^{u} f$; hence $m|u - q_F| \leq |F(u) - \tau| \leq M|u - q_F|$, and integrating over the interval yields $\frac{m}{2}(q - q_F)^2 \leq S(q) - S(q_F) \leq \frac{M}{2}(q - q_F)^2$. $\square$

## Proof of Proposition 1 of the paper (split-conformal validity)

Let $k = \lceil (n+1)\tau \rceil$ and assume $1 \le k \le n$ so that the $k$-th order statistic $e_{(k)} = c_\tau$ of the $n$ calibration errors $\{e_1, \dots, e_n\}$ exists; when $k = n+1$ (i.e. $\tau$ within $1/(n+1)$ of 1) set $c_\tau = +\infty$, giving coverage 1. Suppose the $n+1$ errors $e_1, \dots, e_{n+1}$ (calibration plus the test error $e_{n+1} = z_{n+1} - \widehat{q}(\tau \mid s_{n+1})$) are exchangeable and almost surely distinct. Then the rank $R$ of $e_{n+1}$ among the pooled $n+1$ values is uniformly distributed on $\{1, \dots, n+1\}$. By construction $\widetilde{q}(\tau \mid s_{n+1}) = \widehat{q}(\tau \mid s_{n+1}) + c_\tau$, so

$$\{z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})\} = \{e_{n+1} \le c_\tau\} = \{e_{n+1} \le e_{(k)}\} = \{R \le k\},$$

the middle equality using a.s. distinctness (with ties the middle equality fails; e.g. if all $e_i$ are equal the event has probability 1, which can exceed the stated upper bound). Therefore

$$\Pr\left(z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})\right) = \Pr(R \le k) = \frac{k}{n+1} = \frac{\lceil (n+1)\tau \rceil}{n+1}.$$

Since $(n+1)\tau \le \lceil (n+1)\tau \rceil < (n+1)\tau + 1$, dividing by $n+1$ gives $\tau \le k/(n+1) < \tau + \frac{1}{n+1}$. The probability is over the joint law of calibration and test points; the guarantee is marginal, not conditional on $s_{n+1}$, because a single scalar shift $c_\tau$ pooled across states cannot deliver state-conditional coverage. $\square$

## Proof of Proposition 2 of the paper (tail wedge)

Fix $\nu > 2$, so the $t_\nu$ law has finite variance and its unit-variance version $Z = T_\nu \sqrt{(\nu-2)/\nu}$ is well defined. The $t_\nu$ density is $\sim C_\nu |x|^{-(\nu+1)}$ as $|x| \to \infty$, so the lower tail is regularly varying with index $-\nu$: $\Pr(T_\nu < -x) = \tilde{c}_\nu x^{-\nu}(1 + o(1))$. Unit-variance scaling multiplies the constant, $\Pr(Z < -x) = c_\nu x^{-\nu}(1 + o(1))$ with $c_\nu = \tilde{c}_\nu((\nu-2)/\nu)^{\nu/2}$, and leaves the index $\nu$ unchanged. Writing $U(\tau) = |Q_\nu(\tau)|$ and inverting the regularly varying tail (Embrechts et al., 1997) gives $\tau = c_\nu U(\tau)^{-\nu}(1+o(1))$, hence $U(\tau) = (c_\nu/\tau)^{1/\nu}(1+o(1))$ as $\tau \downarrow 0$, i.e. $Q_\nu(\tau) = -(c_\nu/\tau)^{1/\nu}(1+o(1))$. For $\nu_1 < \nu_2$,

$$\frac{|Q_{\nu_1}(\tau)|}{|Q_{\nu_2}(\tau)|} = \frac{(c_{\nu_1}/\tau)^{1/\nu_1}}{(c_{\nu_2}/\tau)^{1/\nu_2}}(1 + o(1)) = \text{const} \cdot \tau^{-(1/\nu_1 - 1/\nu_2)}(1 + o(1)) \longrightarrow \infty$$

because $1/\nu_1 > 1/\nu_2$. Thus for all sufficiently small $\tau$ both quantiles are negative with $|Q_{\nu_1}(\tau)| > |Q_{\nu_2}(\tau)|$, i.e. $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$. The ordering therefore holds for every $\tau < \delta$ for some $\delta > 0$; putting $\tau^* = \min(\delta, \frac{1}{2})$ gives $\tau^* \in (0, \frac{1}{2})$ as asserted. (This establishes *existence* of $\tau^*$; the actual crossover $\tau_c$ where the ordering flips is not claimed here and is located numerically in the crossover remark of the paper.) Under local truth $t_{\nu_1}$, applying the corresponding equation of the paper with $q = Q_{\nu_2}(\tau)$ and $q_F = Q_{\nu_1}(\tau)$ bounds the regret below by $\frac{m}{2}(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau))^2$, with $m$ the minimum of the $t_{\nu_1}$ density on the wedge interval. $\square$

## References

Paul Embrechts, Claudia Klüppelberg, and Thomas Mikosch. *Modelling Extremal Events for Insurance and Finance*. Springer, 1997.